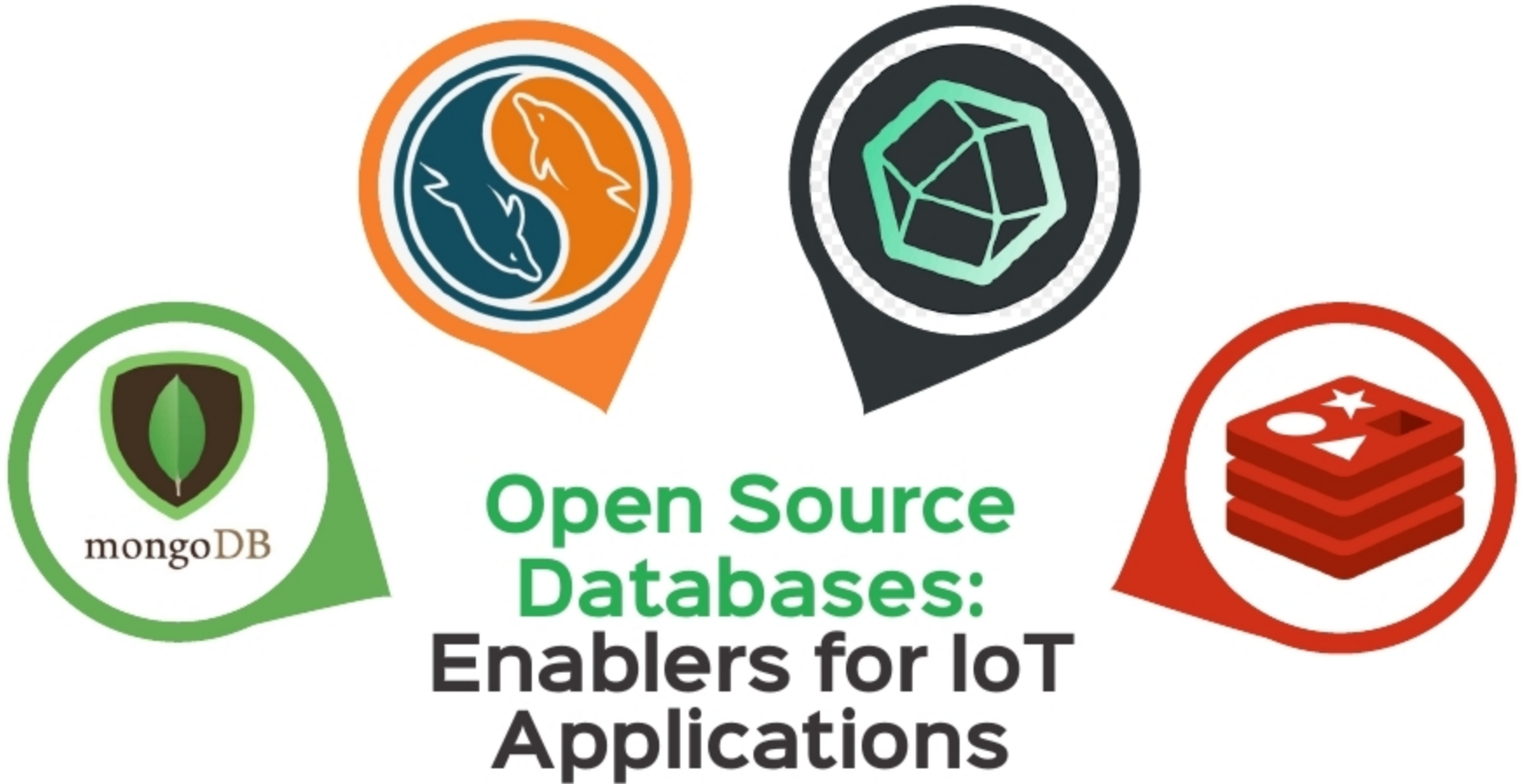




With billions of devices generating trillions of bytes of data, it is crucial for enterprises to be able to organise, store and work with all of it. In this context, databases play a crucial role. The important parameters for choosing the right database for IoT applications are scalability, availability, the ability to handle huge amounts of data, high processing speed and schema flexibility, integration with varied analytical tools, security, and costs.

The Internet of Things (IoT) refers to everyday objects that are readable, recognisable, locatable, addressable and/or controllable via the Internet, irrespective of the communication means — RFID, wireless LAN, wide area networks or any other.

With more physical objects and smart devices connected in the IoT landscape, the impact and value that IT brings to our daily lives becomes stronger. People can make better and more informed decisions when it comes to taking the best routes to work or choosing their favourite restaurants. New services can emerge to address societal challenges such as remote health monitoring for elderly patients.

For enterprises, IoT results in tangible business benefits because it leads to improved management and the tracking of assets and products, new business models, operational

efficiencies, and cost savings due to the optimisation of equipment and resource usage.

According to Statista, the total installed base of IoT connected devices is projected to be 75.44 billion worldwide by 2025. With billions of devices generating trillions of bytes of data, it is crucial for enterprises to be able to organise, store and work with all the data that is generated.

In general, IoT applications leverage both types of databases, relational and non-relational. Non-relational databases are also called NoSQL. IoT requires the functions of both relational and NoSQL databases. The selection of the type of the database is done depending on the type of application. In most of the cases, a combination of both the databases is used.

Most IoT applications are heterogeneous and domain-centric. Choosing the most efficient database for an application can be challenging.

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Gartner Inc. forecasts that by 2020, 20.8 billion connected devices will be in use. Ericsson has predicted that, by 2021, the number of mobile connections worldwide will reach 27.5 billion, including 15.7 billion IoT connections and 8.6 billion mobile phone connections.

Drivers for open source databases for IoT

Most IoT solutions are distributed across geographies. The solutions adopt fog computing at the edge and cloud computing at the enterprise levels. No single database product in the market can fulfil all the needs of the